National College of Ireland
Master of Science in Cloud Computing -MSCCLOUDJAN25AO, MSCCLOUDJAN25BO

Semester One Examination -2025-26
Tuesday 13 th January 2026
6:30pm -9:30pm

Data Governance, Compliance and Ethics

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Written Exam - 60%

Duration: 3 Hours
Maximum Marks: 80

General Instructions:

- This is a compulsory exam. Attempt all parts, including the multiple-choice, short-answer, and long-answer questions.
- The assessment will be **marked out of 80** and will be worth **60%** of the total module marks.
- You have 2 hours to complete this exam. Plan your time carefully to ensure each section receives the appropriate attention.

Section-Specific Instructions

Part 1: Short Answer Questions (30 marks)

Provide clear, concise responses to each question. Focus on addressing the key points directly, supporting your answers with relevant examples or reasoning where indicated. Avoid overly lengthy answers - aim for responses that are direct and relevant to the question.

Part 2: Case Study (35 marks)

Answer all questions in this section. For each question in this section, the response should be provided keeping in mind the company's nature, type of data being handled, location, etc. Read all the information carefully before crafting your response.

Part 3: Multiple-Choice Questions (15 marks)

For each question, mark the correct answer(s) as specified. Some questions may have more than one correct answer - select all that apply. Read each option carefully, as answers that are partially correct will not receive full marks.

Part-1: Short Answer Type Questions - 30 marks

1. Compare centralized, federated, and hybrid data governance models. Which model best suits a multinational enterprise? **[10 marks]**
2. Explain when must a company appoint a Data Protection Officer (DPO) under GDPR? **[6 marks]**
3. A company stores EU customer data on U.S. servers. What steps should it take to ensure compliance with international transfer rules? **[7 marks]**
4. Describe two mechanisms an organization can use to promote transparency in automated systems.

   [7 marks]

Part-2: Case Study/Long Answer Type Questions - 35 marks

Case Study: CloudMed Analytics Ltd.

CloudMed Analytics is a Dublin-based healthcare technology start-up. The company partners with several Irish hospitals to develop predictive models that identify patients at risk of chronic illness. Hospitals share anonymized and pseudonymised patient data with *CloudMed*, which stores the information on a commercial cloud service based in the European Union.
Recently, a journalist raised concerns that *CloudMed* might be using more patient data than necessary, and that patients were not clearly informed about how their data would be used. The management team has asked you, the company's Data Protection Officer (DPO), to prepare a short advisory note outlining the company's key responsibilities and next steps.

1. Outline the main components of an effective data governance framework suitable for *CloudMed*. In your answer, describe the purpose of at least three key roles (such as **Data Owner**, **Data Steward**, **Data Custodian**, or **DPO**) and explain how clear responsibilities can help reduce compliance risks. **[10 marks]**
2. Explain how CloudMed can demonstrate compliance with core GDPR principles such as lawfulness and data minimization. Your answer should mention at least two practical measures. **[10 marks]**
3. Discuss two ethical risks that may arise when using patient data for predictive analytics (e.g bias, transparency, etc.) For each risk, suggest one realistic action *CloudMed* could take - technical or organizational to ensure its analytics remain both ethical and trustworthy. **[15 marks]**

Part-3: Multiple-Choice & Multiple-Select Questions - 15 marks

Select the most appropriate answer(s) for each question. Some questions may have more than one correct option, select all that apply. No partial marks are awarded for incomplete or incorrect answers.

1. Which of the following are core principles of an effective data governance framework? (Select ALL that apply) **[2 marks]**
   a) Clearly defined data ownership and accountability
   b) Minimal documentation of data processes
   c) Data quality, integrity, and provenance
   d) Unrestricted data sharing across departments

2. Under the GDPR, which principle requires organizations to limit the amount of personal data they collect to what is necessary for a specific purpose? **[2 marks]**
   a) Purpose limitation
   b) Data minimization
   c) Integrity and confidentiality
   d) Accuracy

3. Which of the following are essential components of a strong data governance framework? (Select ALL that apply) **[3 marks]**
   a) Clearly defined ownership and stewardship roles
   b) Ad-hoc decision-making with minimal documentation
   c) Data quality, integrity, and lifecycle management
   d) Risk-based access control and classification

4. An AI recruitment system tends to recommend male candidates for technical jobs more often than female candidates with similar profiles. Which ethical principles or fairness concepts are being violated? **[3 marks]**
   a) Group fairness
   b) Transparency
   c) Non-discrimination
   d) Accuracy

5. When conducting a Data Protection Impact Assessment (DPIA), which of the following actions should be taken? (Select ALL that apply) **[3 marks]**
   a) Identify potential privacy risks before processing begins
   b) Document mitigation measures and residual risks
   c) Skip consultation with the DPO to save time
   d) Review processing against GDPR Articles 35 and 36

6. Match each governance role to its key responsibility. **[2 marks]**

Role	**Responsibility**
Data Owner	A. Maintains metadata and daily data quality tasks
Data Steward	**B.** Provides business decisions on data usage
Data Custodian	C. Implements and controls technical storage & access

   Correct pairing options:
   a) 1-B, 2-A, 3-C
   b) 1-C, 2-B, 3-A
   c) 1-A, 2-B, 3-C
   d) 1-B, 2-C, 3-A